

CSCI 4401/5401
Principles of Operating Systems
Fall Semester 2025
Assignment 3

Due Wednesday, October 15th @ 11:59pm

Reading: Tanenbaum Chapter 3

Submission Guidelines:

1. This assignment is worth 30 points for all students.
 - +6 extra credit points if submitted by Tuesday, October 7th @ 11:59pm.
 - +3 extra credit points if submitted by Friday, October 10th @ 11:59pm.
2. Screenshots must use the PrintScreen command so that they are readable.
3. It is your responsibility to make sure all files are readable and submitted on time.
4. Do not include the data files you will download; they will be too large for me to download. You will lose points if you do this.

Submission:

Below is a summary/checklist of what you will submit:

- A main pdf file named "Assignment3.pdf". Your name at top. This will contain:
 - A screenshot for Task 1
 - A screenshot for Task 2
 - A written discussion for Task 3.
- Task 1 – a Java file named "Task1.java". Name as a comment at the top.
- Task 2 – a Java file named "Task2.java". Name as a comment at the top.

Note: we will discuss useful hints in class, e.g., how to iterate over all of the data files in a directory rather than typing them out individually.

Introduction. You will work with a flight prices dataset ([Flight Prices](#)). I modified this dataset so that I could put it on Canvas for you. First, I reduced the size of the dataset; the full dataset is ~32 GB, and I reduced it to ~7 GB. Second, I divided this 7 GB of data across a total of 40 files (each file is ~180 MB). On Canvas, I compressed these files into four groups.

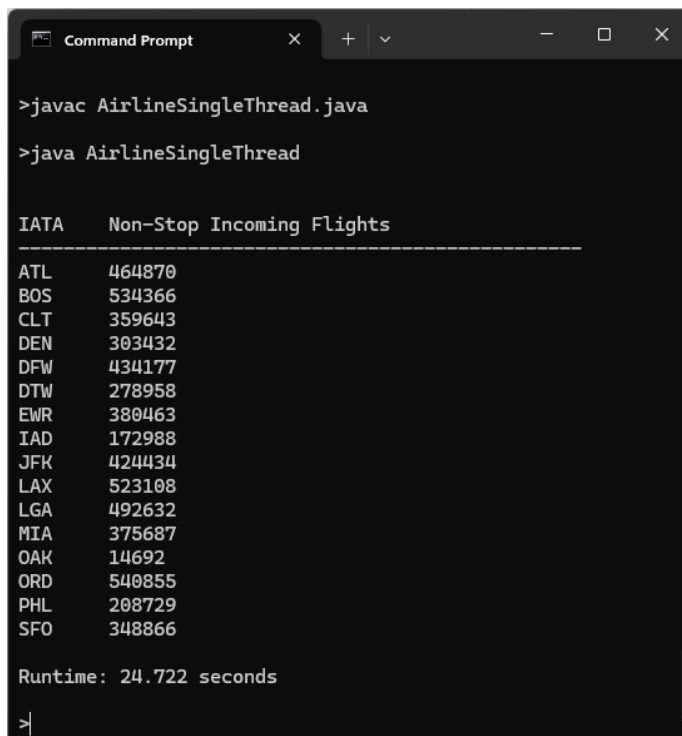
Setup. Download the four .zip files from Canvas: group1.zip, group2.zip, group3.zip, group4.zip. Unzip these and place all .csv files in a single directory (i.e., you will have a directory with a total of 40 .csv files). If you want to view a file, I recommend using a text editor, not a spreadsheet editor like Excel. DO NOT include the data files in your submission; it will be too large.

Task 1: Single Threaded Program (12 points).

Implement a single-threaded Java program that will find the total number of non-stop flights that arrived at each destination airport. Report the runtime for your program. Below is an example screenshot of what your output should look like – your format does not have to exactly match mine. Below are the requirements for your code:

- Print (to the console) the IATA code for each destination airport and the total number of non-stop flights for that destination airport.
- Time your program and print this time to the console.

Submit: 1) Your Java source code in a separate file, and 2) include in your main PDF a screenshot of the output printed to the console.



```
>javac AirlineSingleThread.java
>java AirlineSingleThread

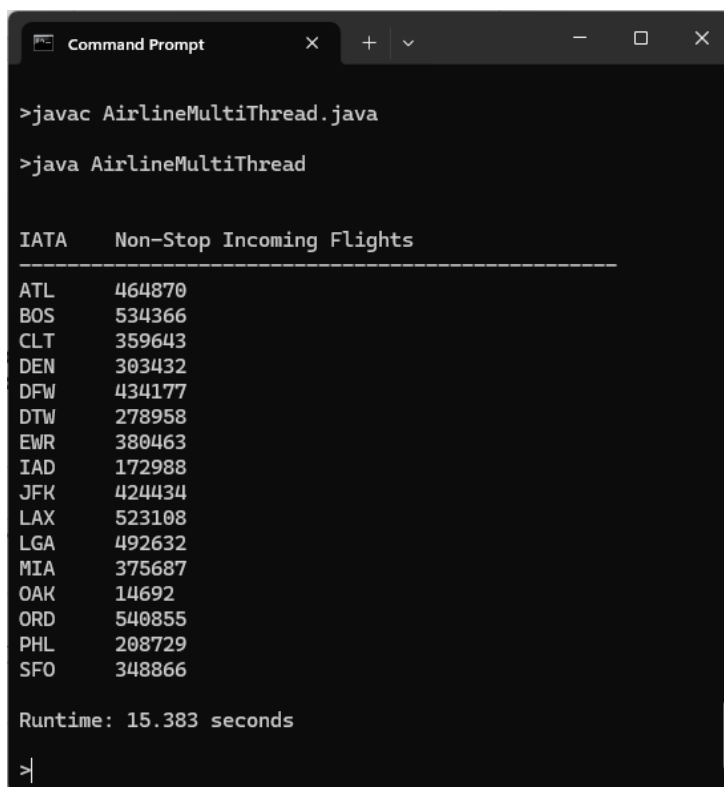
IATA      Non-Stop Incoming Flights
-----
ATL        464870
BOS        534366
CLT        359643
DEN        303432
DFW        434177
DTW        278958
EWR        380463
IAD        172988
JFK        424434
LAX        523108
LGA        492632
MIA        375687
OAK        14692
ORD        540855
PHL        208729
SFO        348866

Runtime: 24.722 seconds
>|
```

Task 2: Multi-Threaded Program (12 points)

Create a new program that modifies your single-threaded program to be multithreaded. It is up to you to decide what code to include in your threads. For example (that you can use), you can create a thread for each file and the thread will find all the non-stop flights for each destination airport. Your new multi-threaded program must be at least 10% faster than your original single-threaded program. For example, if your single-threaded runtime was 100 seconds, the multithreaded program must be 90 seconds or faster.

Submit: 1) Your Java source code in a separate file, and 2) include in your main PDF a screenshot of the output printed to the console.



```
>javac AirlineMultiThread.java
>java AirlineMultiThread

IATA    Non-Stop Incoming Flights
-----
ATL      464870
BOS      534366
CLT      359643
DEN      303432
DFW      434177
DTW      278958
EWR      380463
IAD      172988
JFK      424434
LAX      523108
LGA      492632
MIA      375687
OAK      14692
ORD      540855
PHL      208729
SFO      348866

Runtime: 15.383 seconds
>
```

Task 3: Discussion (6 points).

What was the percentage runtime improvement you were able to achieve? State the improvement of the runtime and describe your approach. This description must clearly describe your approach and explain why you chose this approach. Include in this discussion any other approaches you tried but were not successful. Use key terms we discussed in class. Vague answers will not receive credit.